1.Coding:

```
from decimal import Decimal, ROUND_HALF_UP

BASE_RATE = Decimal("30.00")
FAT_RATE = Decimal("0.40")
SNF_RATE = Decimal("0.30")


def money(value):
    return Decimal(str(value)).quantize(
        Decimal("0.01"), rounding=ROUND_HALF_UP
    )


def rate_per_litre(fat, snf):
    fat_steps = round((float(fat) - 3.5) / 0.1)
    snf_steps = round((float(snf) - 8.5) / 0.1)

    rate = (BASE_RATE +
            fat_steps * FAT_RATE +
            snf_steps * SNF_RATE)

    return money(rate)


def validate_entry(litres, fat, snf):
    if litres < 0:
        raise ValueError("Litres cannot be negative")
    if not 2.0 <= fat <= 7.0:
        raise ValueError("Fat must be between 2.0 and 7.0")
    if snf < 0:
        raise ValueError("SNF cannot be negative")


def payout(litres, fat, snf):
    validate_entry(litres, fat, snf)
    rate = rate_per_litre(fat, snf)
    return money(Decimal(str(litres)) * rate)


def ten_day_total(entries):
    total = Decimal("0.00")

    for litres, fat, snf in entries:
```

```
        total += payout(litres, fat, snf)

    return money(total)

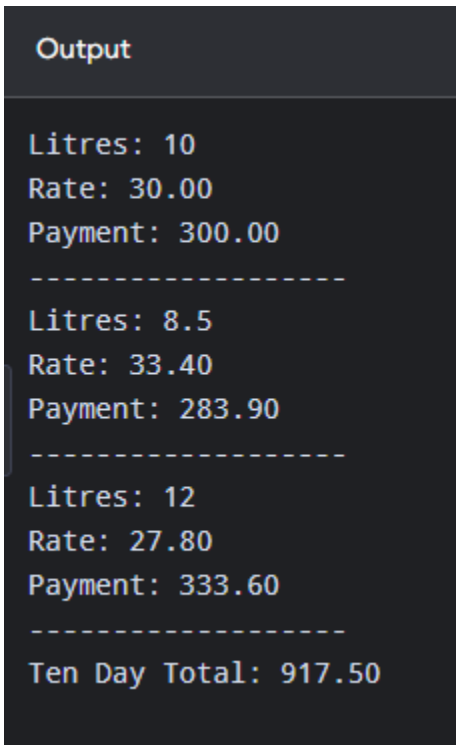

# Main program
entries = [
    (10, 3.5, 8.5),
    (8.5, 4.2, 8.7),
    (12, 3.1, 8.3)
]

for litres, fat, snf in entries:
    rate = rate_per_litre(fat, snf)
    amount = payout(litres, fat, snf)

    print("Litres:", litres)
    print("Rate:", rate)
    print("Payment:", amount)
    print("--------------------")

print("Ten Day Total:", ten_day_total(entries))
```

Output:



2.Coding:

```
from decimal import Decimal, ROUND_HALF_UP

BASE_RATE = Decimal("30.00")
FAT_STEP_VALUE = Decimal("0.40")
SNF_STEP_VALUE = Decimal("0.30")


def money(value):
    return Decimal(str(value)).quantize(
        Decimal("0.01"), rounding=ROUND_HALF_UP
    )


def rate_per_litre(fat, snf):
    fat_steps = round((fat - 3.5) / 0.1)
    snf_steps = round((snf - 8.5) / 0.1)

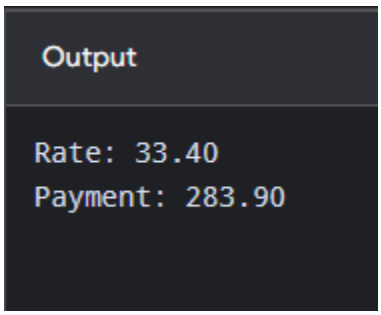
    rate = (BASE_RATE +
            fat_steps * FAT_STEP_VALUE +
            snf_steps * SNF_STEP_VALUE)

    return money(rate)


def payout(litres, fat, snf):
    rate = rate_per_litre(fat, snf)
    return money(Decimal(str(litres)) * rate)


if __name__ == "__main__":
    print("Rate:", rate_per_litre(4.2, 8.7))
    print("Payment:", payout(8.5, 4.2, 8.7))
```

Output:



3.Coding:

```
from decimal import Decimal, ROUND_HALF_UP

# Rate card
BASE_RATE = Decimal("30.00")
FAT_STEP_VALUE = Decimal("0.40")
SNF_STEP_VALUE = Decimal("0.30")


def rate_per_litre(fat, snf):
    fat_steps = int(
        ((Decimal(str(fat)) - Decimal("3.5")) / Decimal("0.1"))
        .quantize(Decimal("1"), rounding=ROUND_HALF_UP)
    )

    snf_steps = int(
        ((Decimal(str(snf)) - Decimal("8.5")) / Decimal("0.1"))
        .quantize(Decimal("1"), rounding=ROUND_HALF_UP)
    )

    rate = (BASE_RATE +
            fat_steps * FAT_STEP_VALUE +
            snf_steps * SNF_STEP_VALUE)

    return rate.quantize(Decimal("0.01"))


def payout(litres, fat, snf):
    rate = rate_per_litre(fat, snf)
    amount = Decimal(str(litres)) * rate
    return amount.quantize(Decimal("0.01"))


# Main program
litres = float(input("Enter milk quantity: "))
fat = float(input("Enter fat percentage: "))
snf = float(input("Enter SNF percentage: "))

rate = rate_per_litre(fat, snf)
amount = payout(litres, fat, snf)

print("\nMILK PAYMENT DETAILS")
print("-------------------------")
```

```
print("Quantity:", litres)
print("Fat:", fat)
print("SNF:", snf)
print("Rate per litre: Rs.", rate)
print("Total payment: Rs.", amount)
```

Output:

